

SAGNIK CHANDRA

MSc in Data Science and AI, RKMVERI, Belur

- +91 6290914274
- Email
- GitHub
- LinkedIn
- Portfolio



SUMMARY

Strong foundation in mathematics, **machine learning**, and deep learning with hands-on experience in **NLP**, **large language models (LLMs)**, **Generative AI (Gen-AI)** pipelines, and distributed **Big Data** processing. Interested in bridging theoretical concepts with practical **AI/ML** systems and building scalable, **data-driven** solutions for real-world **Data Science** applications.

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute

MSc in **Data Science and AI**, 2025–27

University of Calcutta

BSc in Mathematics, 2020–23

CGPA – 7.468

EXPERIENCE

Research Intern – Indian Statistical Institute

May 2026 – Ongoing

- Investigated the “**Lost in the Middle**” phenomenon in **LLM-based RAG** pipelines via Multi-Document QA and Synthetic Key-Value Retrieval on **Llama 3.1 8B Instruct**, plus a threshold-based **attention-sink** analysis.
- Built an end-to-end non-factoid **RAG** evaluation pipeline on **MS MARCO 2.1** dataset for **TREC RAG 2024** benchmark using **BM25** retrieval, **UMBRELA** judgments, and **AutoNuggetizer** to grade **Mistral-7B-Instruct** responses.

Data Science Intern – DeepThought CultureTech Ventures

Oct 2024 – Jul 2025

- Led 10+ cross-functional initiatives spanning **AI automation**, CRM optimization, **data** operations, recruitment, UX research, and business growth.
- Built **AI-powered workflows** and redesigned CRM/KPI reporting, **cutting manual effort by 1–4 hrs/day**, turnaround from **3 days to 1**, and **improving efficiency by 60%** for **30+ stakeholders**.

SKILLS

- Languages:** Python, C, Cypher, R
- Frameworks:** Scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, PySpark, LangChain, LangGraph
- Tools:** Qdrant, Neo4j, Streamlit, Git, GitHub, Jupyter, Optuna, Docker
- Cloud Services:** GCP, AWS

CERTIFICATIONS

- Optimisation Using Genetic Algorithms & Neural Networks** (IIT Kharagpur, 2026)
- Neural Networks and Deep Learning** (Coursera, 2026)

PROJECTS

RAGGY — Adaptive Hybrid RAG Pipeline [GitHub]

Personal Project

Jul 2026 – Ongoing

Built an adaptive **RAG** pipeline orchestrated with **LangGraph**, using a learned query router to dynamically select between **dense**, **sparse (BM25)**, and **RRF**-fused hybrid retrieval over a **Qdrant** vector store (**BGE** embeddings), followed by **cross-encoder** reranking and grounded generation via **Gemini**. Deployment via **GCP** cloud run is currently in progress

Neural Text Style Transfer with Adversarial Learning [GitHub]

RKMVERI

Jan 2026 – May 2026

Built a semi-automated pipeline to **rewrite Bengali sentences in the style of five renowned authors without parallel corpora** using **PyTorch**. Trained a **BiGRU** encoder with a Gradient Reversal Layer and a style-conditioned GRU decoder to disentangle author style, reducing discriminator accuracy from ~65% to near chance (~20%). Curated an 18,779-sentence corpus (Tagore, Sarat Chandra, Satyajit Ray, Bibhutibhushan, Sunil Gangopadhyay) via a **pdfplumber** extraction pipeline with **Tesseract OCR** fallback.

AcademicLens – Citation Graph Mining at Scale [GitHub]

RKMVERI

Jan 2026 – May 2026

Built a large-scale **Big Data** academic intelligence system over 10M+ OpenAlex research papers using **PySpark** and **Neo4j** to analyze research influence, collaborations, and topic trends. Designed a **distributed** ETL pipeline with a multi-hop Author–Paper–Citation–Topic graph and applied **PageRank** for influence ranking.

Stellar Object Classification using Tree-Based Ensembles [GitHub]

RKMVERI

Oct 2025 – Dec 2025

Developed a supervised multi-class classifier for galaxies, quasars, and stars from the SDSS DR18 survey. Evaluated **Random Forest**, **XGBoost**, **LightGBM**, **CatBoost**, and **AdaBoost** under class imbalance, with feature engineering to remove correlated ($|r| > 0.9$) columns and **Optuna** for Bayesian hyperparameter tuning. **CatBoost** and **XGBoost** were top performers, achieving test accuracy above 99% and macro F1 of 0.987.

Traffic-Aware Single-Drone Delivery Route Optimisation [GitHub]

RKMVERI

Oct 2025 – Dec 2025

Implemented **deterministic and stochastic hill climbing** algorithms for a drone delivery route-optimization problem over 120 delivery locations, modelling travel cost via Euclidean distance and node-level congestion as a combinatorial optimization over permutations of delivery nodes. **Deterministic hill climbing achieved the best route (112.36 hrs)** and converged faster but was more sensitive to initial configuration; **stochastic hill climbing traded convergence speed for broader search-space exploration (138.90 hrs)**.